

Ultrasonic Echo-Based Material Classification System

Comprehensive Parameter Design, Mathematical Formulation & Firmware Evaluation

PART I: Parameters Required for Strong Material Classification

1. Introduction

Material classification using ultrasonic echoes depends on how different materials reflect, absorb, and distort acoustic energy. These behaviors depend on acoustic impedance ($Z = \rho c$), density, elasticity, porosity, and surface roughness. Robust classification requires physically meaningful and distance-invariant features.

2. Time-Domain Features

- Peak Amplitude ($A_{max} = \max|x[n]|$): Hard materials reflect strongly. Must be normalized by distance squared.
- Echo Width: Measured at 20%, 50%, and 80% thresholds. Hard materials produce narrow pulses; soft materials produce wider pulses.
- Rise Time: Time between 10% and 90% amplitude. Indicates surface rigidity.

3. Envelope Features

Proper envelope: $E[n] = \sqrt{x[n]^2 + \text{Hilbert}(x[n])^2}$.

- Envelope Energy: $\sum(E[n]^2)$. Indicates absorption.
- Envelope Area: $\sum(E[n])$. Must be distance normalized.
- Multi-threshold widths improve shape discrimination.

4. Frequency-Domain Features (Critical)

After FFT:

- Spectral Centroid: Weighted frequency mean.
- Spectral Bandwidth: Spread of frequency content.
- Dominant Frequency.
- High/Low Frequency Energy Ratio (captures attenuation behavior).

5. Statistical Shape Features

- Kurtosis: Detects sharp metal reflections.
- Skewness: Detects asymmetry in pulse shape.

6. Chirp-Based Features

True matched filter: Cross-correlation with transmitted chirp.

- Compressed peak amplitude.
- Compression width.
- Side-lobe ratio.
- Frequency-dependent attenuation curve.

7. Distance-Invariant Normalization

Amplitude decays as $1/r^2$.

Corrected amplitude = $A * r^2$.

Without normalization, classifier learns distance instead of material.

Recommended Strong Feature Vector:

[Peak_norm, Energy_norm, Width_50, Width_20, Rise_time, Spectral_centroid, Spectral_bandwidth, High_low_ratio, Kurtosis, Skewness]

PART II: Critical Review of Current STM32 Firmware

1. Incorrect Matched Filter

The implemented filter is a second-order IIR resonator, not a true matched filter. A real matched filter requires cross-correlation with stored chirp samples. Without correlation, pulse compression and frequency discrimination are lost.

2. Improper Envelope Extraction

Envelope is implemented as raw magnitude only. No Hilbert transform or analytic signal is used. This reduces classification precision.

3. Area Not Normalized

Area under pulse depends on distance ($1/r^2$ decay). Without normalization, classifier confuses distance with material.

4. RMSE/MAE Misinterpretation

RMSE is computed relative to moving mean, not ground truth. This measures stability, not accuracy. Accuracy_percent is mathematically invalid.

5. Power Proxy

Power_percent is duty cycle estimation, not real power measurement. It does not contribute to material classification.

6. Missing Frequency-Domain Features

No FFT, spectral centroid, bandwidth, or frequency energy ratios are computed. These are critical for separating materials like metal vs glass.

7. No Feature Vector or Classifier

The firmware outputs distance-related metrics but lacks a discriminative feature vector and does not implement any classification algorithm.

Conclusion:

The current system is an advanced ultrasonic ranging platform but not a material classification system. To enable classification, true matched filtering, FFT-based feature extraction, distance normalization, statistical shape features, and a trained classifier must be added.